Solar Energy 241 (2022) 553–563

Contents lists available at ScienceDirect

Solar Energy

journal homepage: www.elsevier.com/locate/solener

A novel water-free cleaning robot for dust removal from distributed photovoltaic (PV) in water-scarce areas

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ARTICLE INFO

Keywords:
Distributed photovoltaic
Water-free
Cleaning robot
Water-scarce
Negative pressure adsorption

ABSTRACT

Distributed photovoltaic (PV) power stations are installed in high-elevation locations and various configurations. Traditional manual cleaning methods suffer from low cleaning quality, low efficiency, and high water consumption, making it difficult to achieve consistent cleaning. This paper proposes a novel water-free cleaning robot for dust removal from PV panels of distributed PV systems in water-scarce areas. A force analysis is conducted, and safe working conditions are considered to ensure the robot can safely clean PV panels installed at a large angle. The material, structure, and sealing device of the robot are optimized based on the carrying capacity of the PV system and the operating efficiency. A combined dust removal system consisting of a rolling brush and negative pressure is developed to prevent dust raising during cleaning. Experiments are carried out on a 2-kW distributed PV system on the roof of a university in Northeast China to verify the effectiveness of the negative pressure adsorption system and the obstacle crossing and cleaning abilities of the robot. The results show that the water-free cleaning robot can effectively remove dust from the panels. The average dust cleaning rate is 92.46%, and the increase rate of the PV efficiency ranges from 11.06% to 49.53%. In addition, the robot has a small volume and weight and is more suitable than manual or mechanical cleaning for dust removal from PV panels of distributed PV systems in water-scarce areas.

1. Introduction

In China, distributed photovoltaic (PV) power stations are typically constructed in remote and water-scarce areas with high solar radiation (Fares et al., 2021; Boddupalli et al., 2017). Dust accumulation affects the efficiency of PV power generation (Chen et al., 2020; Alnasser et al., 2020). According to statistics, the power generation efficiency of PV panels decreases by about 26% when the dust concentration reaches 22 g/m^2 (Jiang et al., 2011). Dust accumulation also changes the thermal balance of PV panels, causing local heating. In serious cases, it will produce hot-spot effects and threaten the safety of the PV generation system (Abderrezek and Fathi, 2017; Tang et al., 2021). PV power stations require regular manual cleaning to remove dust from the panels. Due to the location, water resource limitation, and the cost of the PV system, manual cleaning is labor-intensive, costly, and inefficient, the water consumption is high, and the cleaning cycle is long. This method is not applicable to distributed PV system (Ullah et al., 2020). It is a novel development trend that the robot replaces human to clean the dust on PV panels (Zhou and Li, 2018; Vega-Heredia et al., 2019; Shehri et al., 2016; Michele et al., 2020). Therefore, it is crucial to develop a water-free cleaning robot to remove dust in a timely and effective manner to reduce cleaning cost and improve the safety of distributed PV systems in water-scarce areas.

Dust accumulates over time on the surface of PV panels. The output power of the PV panels depends on the solar radiation energy, and dust accumulation on the panel surfaces reduces the absorption of energy and the photoelectric conversion efficiency, resulting in an output loss of the PV system of 2%–10% or up to 25% in serious cases (Monto and Rohit, 2010; Majeed et al., 2020; Andrea et al., 2019). Several researchers investigated the influence of dust accumulation on the performance loss of PV power generation systems. Klugmann-Radziemska (2015) carried out a natural dust accumulation experiment of PV panels in Poland. The results showed that 1 mm of dust reduced the conversion efficiency by 25.5% and the maximum power by 3% on average. Hachicha et al. (2019) measured the influence of dust accumulation on the PV output performance using indoor and outdoor experiments in the United Arab Emirates. The results showed that the dust concentration had a linear relationship with the normalized PV power, and the PV power decreased by 1.7% / g/m^2. The reliability of the linear relationship was verified by outdoor experiments. The PV

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https://doi.org/10.1016/j.solener.2022.06.024
Received 9 February 2022; Received in revised form 17 May 2022; Accepted 15 June 2022
Available online 1 July 2022
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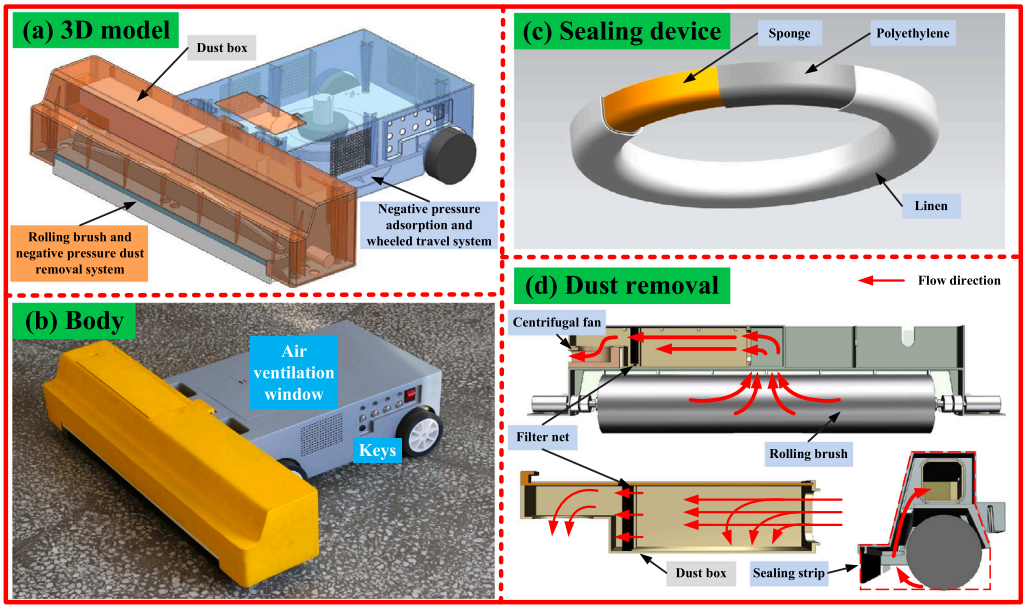
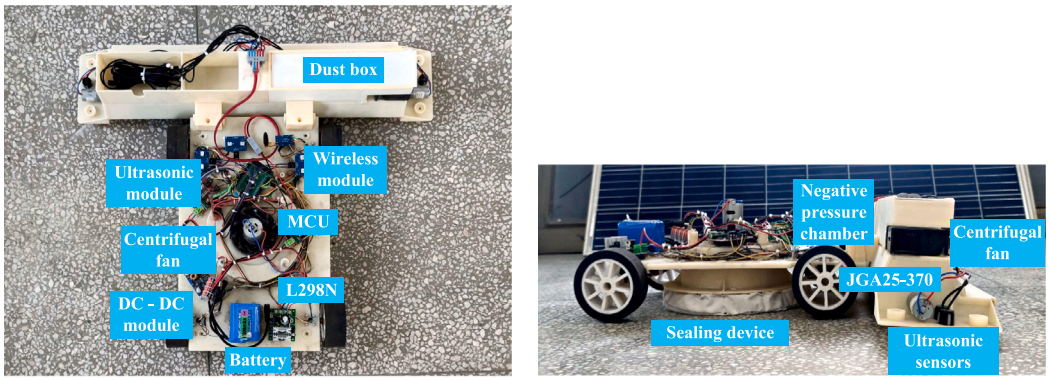


Fig. 1. The water-free cleaning robot for PV distributed system.

power loss caused by dust accumulation was 12.7%, and dust concentration increased by 5.44 g/m^2. Paudyal and Shakya (2016) established a regression equation to describe the relationship between the dust concentration, meteorological variables, and conversion efficiency loss of PV panels. During the 5-month study period, the dust concentration on the PV panels was 9.6711 g/m^2, and the conversion efficiency of the dusty PV panels was 29.76% lower than that of panels that were cleaned daily. In addition, dust particles accumulated primarily at the bottom of the PV panels, resulting in "hot spots" and other risks, which could cause permanent damage to the PV panels. Salari and Hakkaki-Fard (2019) studied the influence of dust concentration on the performance of PV panels using numerical simulation methods. The results showed that the conversion efficiency of the PV panels decreased by 26.36% as the dust concentration increased from 0 g/m^2 to 8 g/m^2. Chen et al. (2020) developed an online monitoring system for meteorological parameters and PV performance to determine the relationship between PV power output attenuation and dust accumulation. The results showed that the dust accumulation on the surface of the PV panels tended to accumulate during rainfall events, which was a primary factor in the rapid decline of the PV output. In East China, the average dust concentration on PV panels in one week was 0.644 g/m^2, resulting in a 7.4% reduction in PV output power. Generally, rainfall is a natural cleaning mechanism for PV panels, although its effectiveness depends on the rainfall level. When it is low, the PV panels cannot be cleaned entirely, and residual dust remains on the PV panels (Kimber et al., 2006). Unfortunately, distributed PV power stations are typically built in low-precipitation and water-scarce areas, where PV cleaning by rainfall is unreliable, resulting in greater surface dust accumulation than in areas with more precipitation.

Current PV cleaning methods include (Kazem et al., 2020) water cleaning, manual cleaning, mechanical cleaning, electrical screens (EDS), superhydrophobic coating, super water jet, and ultrasonic cleaning. Researchers developed PV cleaning equipment based on these methods. Inspired by the lotus leaf effect, superhydrophobic coating suitable for PV components were prepared (Sutha et al., 2017; Gwon et al., 2014; Roslizar et al., 2019). The surfaces of PV panels were covered with films with excellent mechanical stability. Rain or fog drops minimized dust accumulation, reducing the need for traditional cleaning, water resources, and labor costs. This method requires sufficient rain for cleaning the panels and is not suitable for distributed PV power stations in arid and water-deficient areas. Meanwhile, coatings that remain on panels for a long time in a harsh environment have a shorter life cycle, which increases the operation and maintenance costs of PV power stations. NASA used EDS to solve for dust removal from spacecraft PV panels (Tatom et al., 1967). Mazumder et al. (2017) reviewed the literature on EDS with a focus on energy consumption and water-free cleaning. Zhao et al. (2019) designed a mobile dry cleaning device for PV panels that used rotating brushes and a vacuum system. The device exhibited good performance in large ground-based PV power stations. These methods are complex and not suitable for distributed PV power stations with low capacity. Parrott et al. (2018) designed a dry-cleaning robot for PV panels and proposed a new type of brush with a silicone foam flap mounted on an aluminum core. This system minimized the impact of dust on the PV panels' power output. Hammoud et al. (2019) designed a cleaning robot with wheels and verified the cleaning performance. The cleaning robot substantially reduced the influence of dust on the power output of PV panels; the power generation increased by 32.27% on average after cleaning. Cai et al. (2019) designed a novel PV cleaning robot. The relationship between the pressure distribution and the dust particle velocity was simulated using computational fluid dynamics. The structural parameters of the suction inlet of the robot were optimized by an orthogonal experiment. The experimental results obtained from the optimized structure were in good agreement with the simulation results, and the average suction speed at the inlet was 41.8 m/s. Although robots have been successfully used for PV cleaning in the literature, they have some limitations. Robots are not suitable for PV panels with a high angle, and some operate on a track; thus, their use is limited by the installation angle and configuration of the PV panels. The above robots do not meet the functional requirements of portability, crawling, dealing with obstacles, and anhydrous cleaning; thus, they are not suitable for dust cleaning of distributed PV systems in water-scarce areas.

This paper proposes a novel water-free cleaning robot for PV panel dust cleaning of distributed PV systems in water-scarce areas. A force analysis of the moving wheels of the cleaning robot is conducted to determine the minimum safe negative pressure required for the robot to operate on the tilted PV panels. A theoretical derivation of the conditions required for the robot's adherence to the panel is performed. A dust removal system using a rolling brush and negative pressure is developed to prevent the raising of dust during cleaning. The bearing capacity and operating efficiency of the PV system are considered to



(a) Top view (b) Side view

Fig. 2. The internal device of the water-free cleaning robot.

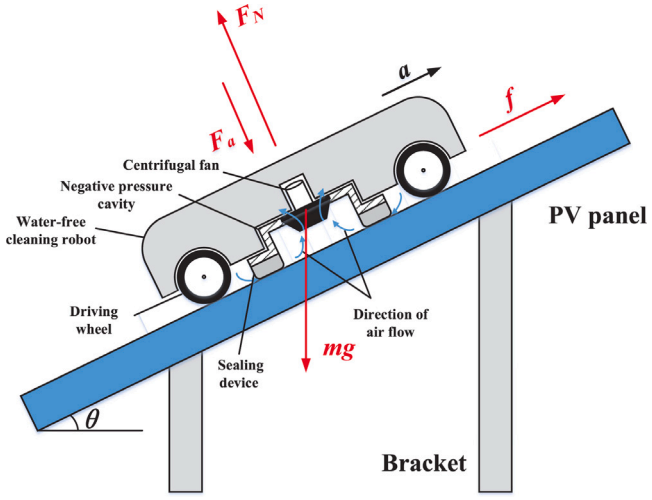


Fig. 3. The mechanical equilibrium model of the robot.

optimize the material, structure, sealing device, and other key parameters of the robot to ensure broad applicability of the robot to different scenarios. Experiments are conducted to determine the negative pressure adsorption, obstacle crossing, and cleaning abilities of the robot to verify its flexibility and reliability. The results provide technical support for the intelligent operation and maintenance of distributed PV systems.

The main contributions of this paper can be summarized as follows:

- A novel water-free cleaning robot is proposed for the dust removal from PV panels of distributed PV systems in water-scarce areas.
- The operating mechanism of the wheeled robot is based on negative pressure adsorption to enable its operation on PV panels with an angle of up to 30°.
- The dust collecting method consisting of a rolling brush and negative pressure prevents dust raising during cleaning.

2. Case description

Distributed PV power stations are typically located in remote areas, such as mountains and hills, abandoned mining areas, and factory roofs. Thus, there are strict requirements on the weight, climbing ability, traveling mechanism, and cleaning method of the robot. The robot's mass cannot be too large, and it should have sufficient cruising ability to enable its operation in hilly areas, on household roofs, and in industrial and commercial workshops. The robot travels on wheels and utilizes negative pressure adsorption. A negative pressure adsorption system

Table 1
The design parameters of the water-free cleaning robot.

Parameters	Value
Weight	⩽8 kg
Dust cleaning rate	⩾90%
Maximum climbing angle	30°
Obstacle height	⩽20 mm
Cruising ability	⩾200 m
Cleaning method	Water-free cleaning

and four independent wheels are used to ensure free movement on the PV panel within 30°. The cleaning system consists of a rolling brush and negative pressure dust removal, ensuring efficient cleaning while preventing the creation of secondary pollution. The design parameters of the water-free cleaning robot are listed in Table 1.

2.1. Robot design

The water-free cleaning robot for the distributed PV system is shown in Fig. 1. It consists of the negative pressure adsorption and wheeled travel system and the rolling brush and negative pressure dust removal system. A high-speed centrifugal fan is used to generate negative pressure. The wheels are powered by 60 rpm/JGA 25-370 motors that control the speed. The Φ 90 soft silica gel wheels provide sufficient friction. The 3D model and the robot prototype are shown in Fig. 1(a) and (b), respectively.

The sealing device is circular and consists of a low-density sponge, polyethylene, and linen, enabling the robot to adhere to the surface of the PV panel during movement, as shown in Fig. 1(c). The dust collection principle is shown in Fig. 1(d). Two 220 rpm motors drive the roller brush, which raises the dust. The centrifugal fan sucks and suspends the dust, which is collected in the dust box. The filtered clean air is discharged to complete the negative pressure dust collection process. The dust box has a size of 160 mm × 67 mm × 58 mm and can store 600 g of dust. It is cleaned manually after cleaning a PV array. Two views of the internal components of the water-free cleaning robot are shown in Fig. 2.

Pulse width modulation (PWM) with switching frequency of 14 kHz is used to control the motor's output to prevent vibration of the centrifugal fans and keep the speed below the maximum value. In addition, an air fairing device is attached to the shell to disperse the incoming air evenly and reduce the airflow inside the robot.

2.2. Safe operating conditions for the robot

The water-free cleaning robot operates on tilted PV panels based on negative pressure adsorption, which prevents sliding. Static force analysis of the robot is carried out to determine the minimum conditions

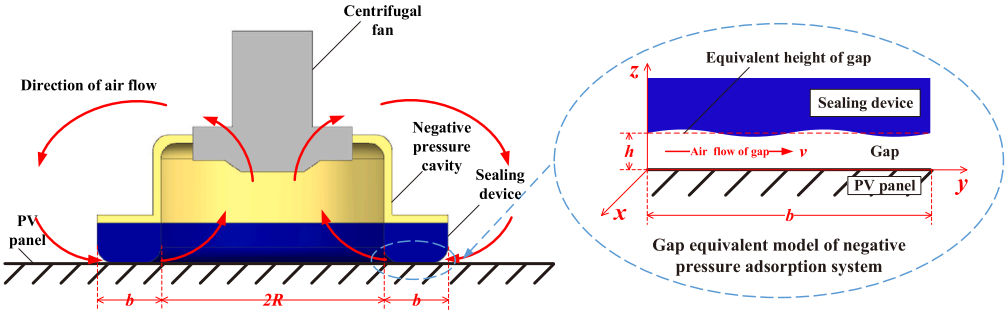


Fig. 4. The principle of the negative pressure adsorption system.

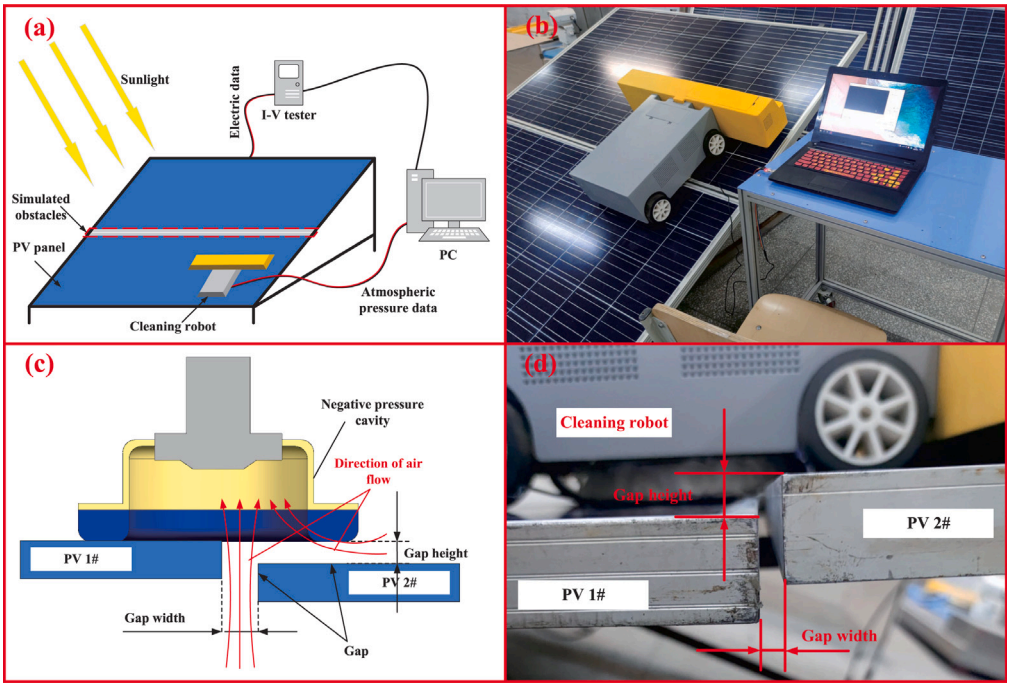


Fig. 5. The experimental system for evaluating the cleaning robot's performance.

required for safe operation. The mechanical equilibrium model of the robot is shown in Fig. 3.

The inclination angle of the PV panels ranges from 0° to 30°. The robot is subjected to gravity $G = mg$, the supporting force F_n, adsorption force F_a, and friction force f. Under the combined action of these forces, the robot moves along the panel with acceleration a. According to the D' Alemberts principle (Wang and Zu, 2017),

$$\mu mg\cos\theta + \mu F_a = mg\sin\theta + ma \tag{1}$$

where μ denotes the friction coefficient between the robot and the PV panel, F_a denotes the adsorption force of the robot, and m denotes the mass of the robot. g denotes the acceleration of gravity, which is 9.8 m/s^2.

Let the ratio of the acceleration a of the robot to the acceleration of gravity g be k. Then, the critical adsorption force F_a required for the safe operation can be described as follows:

$$F_a = \frac{mg}{\mu}(\sin\theta - \mu\cos\theta + k) \tag{2}$$
$$= \frac{mg}{\mu}\left[\sqrt{1+\mu^2}\sin(\theta - \arctan\mu) + k\right]$$

If $F_a = 0$, the critical angle of the adsorption force $\theta_{nt} = \arcsin\left(-k/\sqrt{1+\mu^2}\right) + \arctan\mu$. θ_{nt} is the angle threshold beyond which the robot requires the adsorption force to ensure the safe operation on the PV panels.

(1) $0° \leqslant \theta \leqslant \theta_{nt}$

At this time, the safe operation of the robot is ensured without relying on the negative pressure adsorption system.

(2) $\theta_{nt} \leqslant \theta \leqslant 90°$

When the robot operates on a PV panel with inclination angle θ, the cleaning robot requires the additional adsorption force provided by the negative pressure adsorption system to ensure safe operation.

3. Adsorption system and field experiment

The negative pressure adsorption system is a key component that ensures the free movement of the water-free cleaning robot on the PV panels, but the high energy consumption affects the working efficiency. The energy efficiency during the safe operation is analyzed. A relationship model between the power consumption and negative pressure is established to determine the critical negative pressure and the power requirements of the centrifugal fan.

3.1. The principle of the negative pressure adsorption system

The airflow direction when the water-free cleaning robot is operating is indicated by the red arrows in Fig. 4. The impeller of the centrifugal fan causes high-speed airflow. The air is discharged from the negative pressure cavity. External air is drawn into the negative

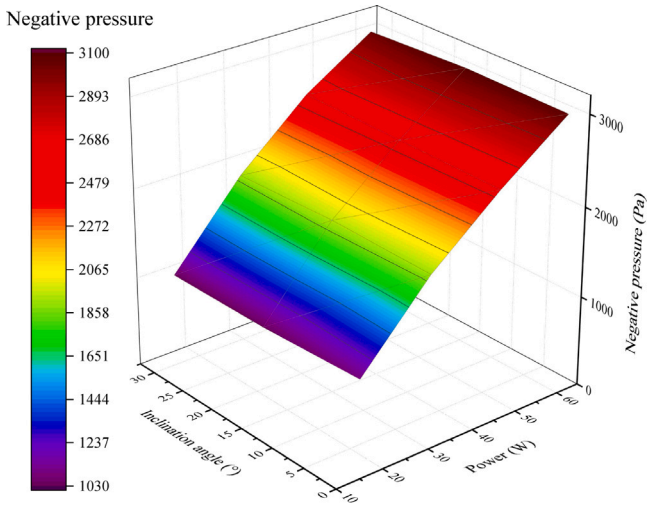


Fig. 6. The relationship between the inclination angle and negative pressure at a constant power of the centrifugal fan.

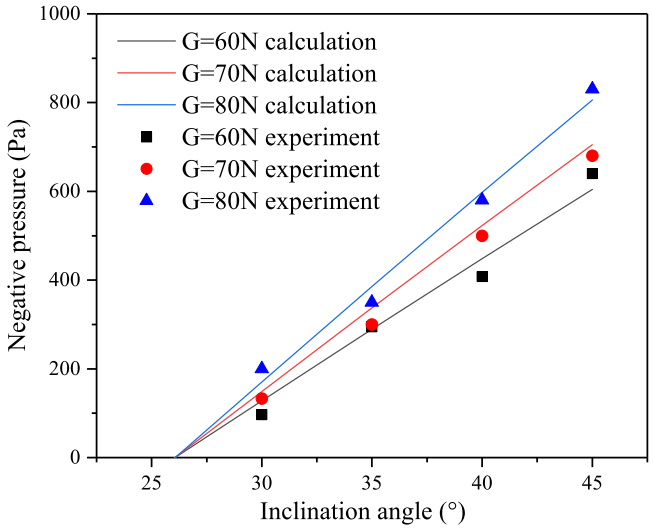


Fig. 7. The deviation between the theoretical and the experimental critical negative pressure for different installation angles of the PV panels.

pressure cavity through the gap between the sealing device and the PV panel due to the pressure difference between the inside and outside of the cavity. Thus, the air circulates, generating stable negative pressure. The leakage air is simplified as airflow between the panels; the equivalent model is shown in the blue box in Fig. 4.

The movement of the leakage air affects the sealing performance of the negative pressure adsorption system. In an atmospheric environment, the adsorption force F_a of the negative pressure adsorption system can be described as:

$$F_a = l\int_0^b (p_0 - p)dx = l\int_0^b \Delta p dx = \Delta p \pi R^2 \tag{3}$$

where Δp denotes the differential pressure on both sides of the sealing device, p_0 denotes the standard atmospheric pressure, p denotes the internal pressure of sealing device, b denotes the sealing device width, S denotes the effective adsorption area of the sealing device, and R denotes the equivalent radius of the effective adsorption area.

When the negative pressure adsorption system reaches a stable state, the air discharged by the centrifugal fan is equal to the air inflow through the gap between the sealing device and the PV panel. The Reynolds number can be obtained as follows:

$$Re = \frac{\rho v h}{\tau} \tag{4}$$

where ρ denotes the air density, v denotes the average velocity of the air, h denotes the equivalent height of the gap between the sealing device and the PV panel, and τ denotes the viscosity of the air.

The height of the leakage gap is small, and the flow rate of the leakage air is slow. The Reynolds number Re of the leakage airflow between the sealing device and the PV panel is about 200 (obtained from Eq. (4)). Therefore, the leakage air has laminar flow. When the negative pressure adsorption system operates in a stable state, the flow velocity is uniform, and the mass point does not change, meeting the conditions of the continuity equation, Navier–Stokes equation, and the flow of an incompressible fluid. The movement inertia force and mass force can be ignored (Jayanthi and Kavitha, 2019). Thus, the average velocity of the leakage air v is:

$$v = \frac{z(h-z)}{2\tau b}\Delta p \tag{5}$$

The flow rate q_v of the leakage air can be expressed as:

$$q_v = \int_0^h v l dz = \frac{l h^3}{12\tau b}\Delta p \tag{6}$$

where $l = \pi(b + 2R)$ denotes the circumference of the circular sealing device. The airflow rate discharged from the pressure chamber by the centrifugal fan is equal to the flow rate q_v of the leakage air. Then, the power consumption P of the centrifugal fan is:

$$P = \Delta p \times q_v = \frac{\pi h^3 (b+2R)}{12\tau b}\Delta p^2 \tag{7}$$

We substitute the critical adsorption force F_a from Eq. (2) into Eq. (3) and (7). The critical negative pressure Δp and the critical power P of the negative pressure adsorption system are:

$$\Delta p = \frac{mg}{\pi \mu R^2}\left[\sqrt{1+\mu^2}\sin(\theta - \arctan\mu) + k\right] \tag{8}$$

$$P = \frac{m^2 g^2 h^3 (b+2R)}{12\pi\tau b \mu^2 R^4}\left[\sqrt{1+\mu^2}\sin(\theta - \arctan\mu) + k\right]^2 \tag{9}$$

Hence, Eqs. (2), (8), and (9) define the critical adsorption force F_a, critical pressure Δp, and critical power P of the negative pressure adsorption system.

3.2. The experimental system

The experimental system for evaluating the water-free cleaning robot's performance is shown in Fig. 5(a) and (b). It includes two CSUN260-60P PV panels, an adjustable bracket (0° to 60°), a BMP280 pressure sensor, PROVA 1011 I-V tester, signal lines, and a computer. The pressure inside and outside of the negative pressure adsorption system is measured by the BMP280 pressure sensor to determine the effects of different centrifugal fan power levels, inclination angles, and gap sizes on the pressure (Fig. 5(c)).

Since the PV panels are installed on variable terrain, they are height differences between them, and a gap exists between two PV panels in the horizontal direction (Fig. 5(d)). When the robot crosses the gap, a large amount of air enters the negative pressure adsorption system, disrupting the negative pressure and adherence of the robot to the panels. The sealing device adopts low density sponge, polyethylene, and linen to improve the sealing performance of the negative pressure adsorption system. By increasing the power of the centrifugal fan and the elastic deformation of the sealing device, the robot can successfully cross the gap.

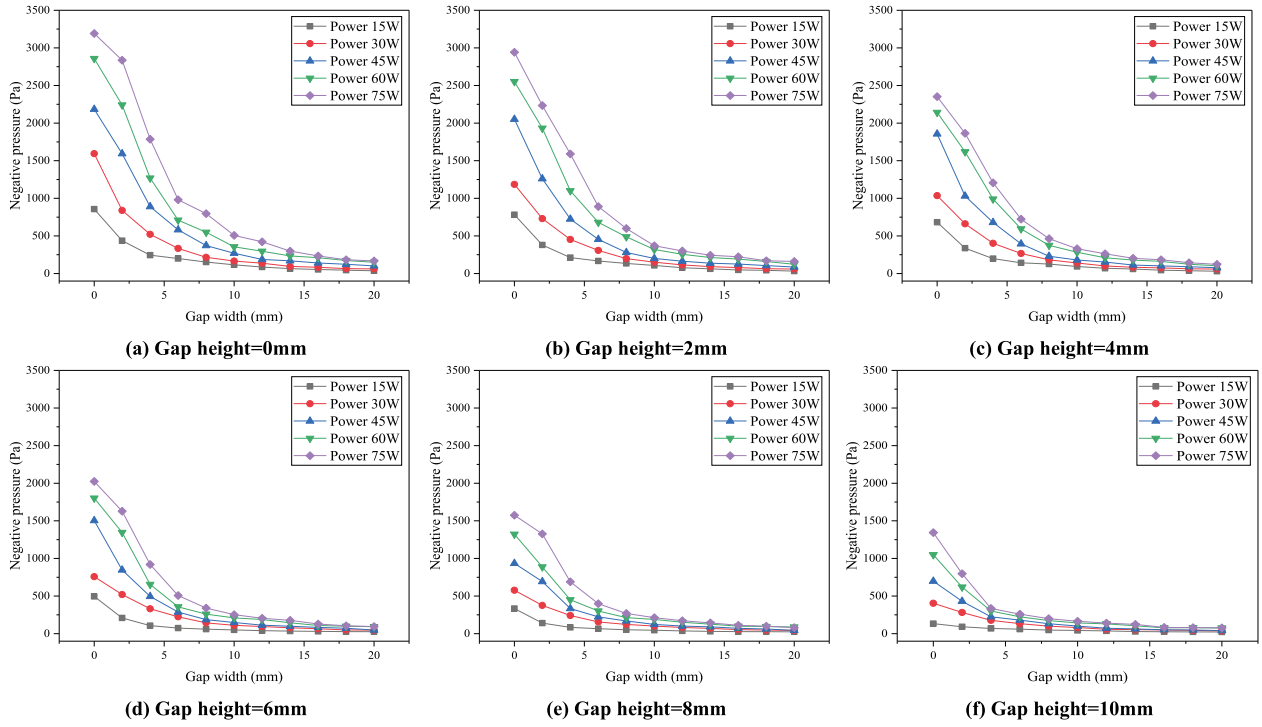


Fig. 8. The relationship between gap the width and negative pressure at the same gap height.

4. Results

4.1. Relationship between inclination angle and negative pressure

The relationship between the inclination angle of the PV panels (0° to 30°) and the negative pressure of the negative pressure adsorption system of the robot at a constant power of the centrifugal fan is shown in Fig. 6.

The results indicate that the inclination angle in the range of 0° to 30° does not affect the negative pressure of the cleaning robot.

The deviation between the theoretical and the experimental critical negative pressure for different installation angles of the PV panels and different load weights is shown in Fig. 7.

An excellent fit is obtained between the critical negative pressure obtained from the experiments and the theoretical calculations. When the other conditions remain unchanged, the critical negative pressure has an approximately sine relationship with the inclination angle. Thus, wheels with a large friction coefficient should be selected to minimize the power consumption of the centrifugal fan.

4.2. Relationship between the gap width and negative pressure

The gap width between the panels ranges from 0 mm to 20 mm. The working state of the negative pressure adsorption system is affected by the gap width. The relationship between the gap width and the negative pressure is evaluated at different power levels of the centrifugal fan (15 W, 30 W, 45 W, 60 W, and 75 W). The experimental results are shown in Fig. 8.

The power of the centrifugal fan, the gap width, and the gap height significantly affect the negative pressure. The negative pressure is 3191 Pa at a gap and height of 0 mm and 75 W power of the centrifugal fan. When the power of the centrifugal fan is 45 W, the negative pressure is 31.56% lower (2184 Pa). The negative pressure is 1856 Pa at 45 W power of the centrifugal fan, a gap width of 0 mm, and a gap height of 4 mm. The negative pressure stabilizes at 394 Pa at a gap width of 6 mm. In this condition, the negative pressure is 78.77% lower than in the gap width of 0 mm.

A change in the gap width has a substantial effect on the negative pressure. When the water-free cleaning robot crosses the gap, sufficient negative pressure is maintained to ensure that the robot adheres to the panel. The robots can cross the gap between the PV panels at lower inclination angles. However, at higher angles, it is necessary to increase the power of the centrifugal fan to increase the negative pressure and ensure sufficient adherence of the robot to the PV panels.

4.3. Relationship between the gap height and negative pressure

The relationship between the gap height (0 mm, 2 mm, 4 mm, 6 mm, 8 mm, and 10 mm) and the negative pressure is shown in Fig. 9.

The negative pressure exhibits a decreasing trend. When the gap width and the power of the centrifugal fan remain unchanged, the negative pressure is linearly related to the gap height. When the gap width is small, it has a large effect on the negative pressure and vice versa.

We use the example of centrifugal fan power of 30 W, and a gap width of 2 mm and 12 mm as an example. As the gap height changes from 0 mm to 10 mm, the negative pressure decreases from 839 Pa to 282 Pa, a 66.39% decrease. When the gap width is 12 mm, and the other conditions remain unchanged, the negative pressure decreases from 137 Pa to 58 Pa, a 57.66% decrease. The reason is that the sealing device of the negative pressure adsorption system stretches from the compression state and fills the gap when the gap height is small. At this time, the sealing performance of the negative pressure adsorption system is high. In contrast, when the gap height is large, the sealing device cannot fill the gap because it cannot stretch sufficiently. Therefore, the sealing performance is lower at a larger gap height, and a large amount of air flows into the negative pressure adsorption system, reducing the negative pressure. A low-density sponge, polyethylene, and linen were used for the sealing device to prevent negative pressure failure and insufficient adherence to the panel. Thus, the water-free cleaning robot can maintain sufficient negative pressure to adhere to the panels.

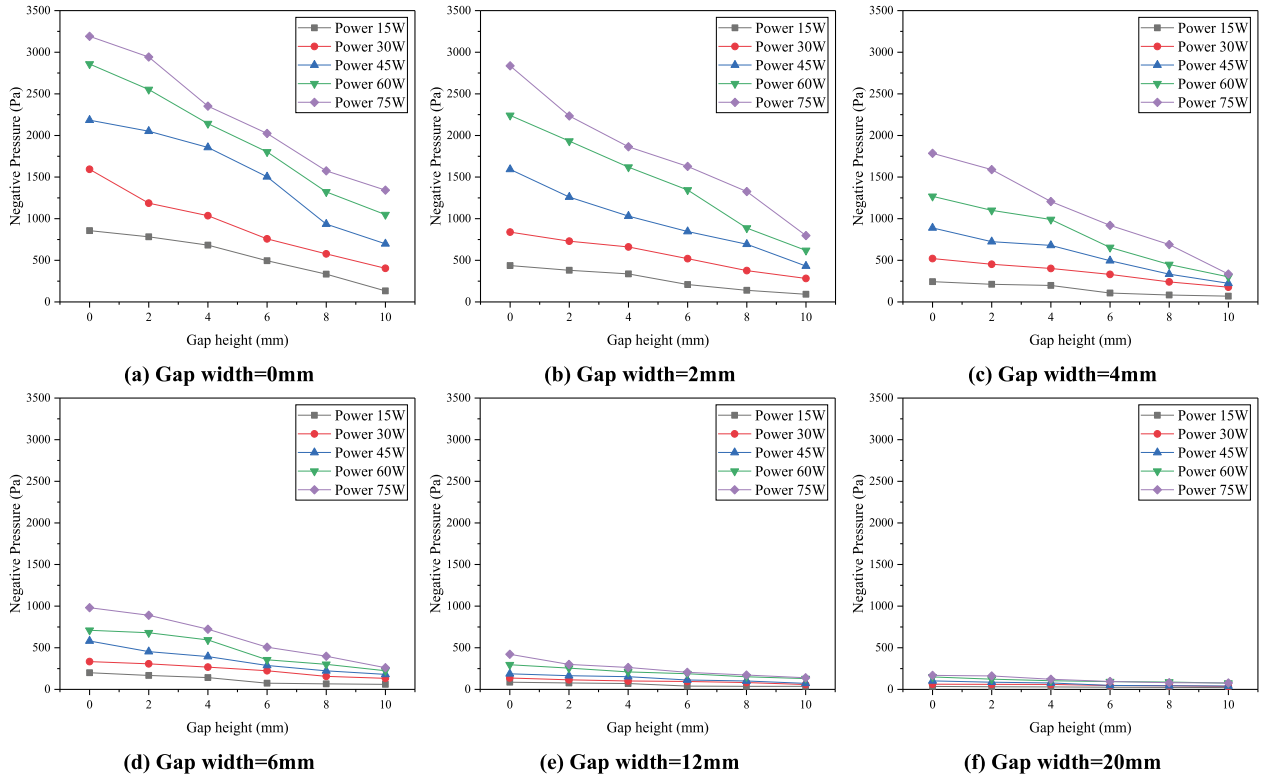


Fig. 9. The relationship between the gap height and negative pressure at the same gap width.

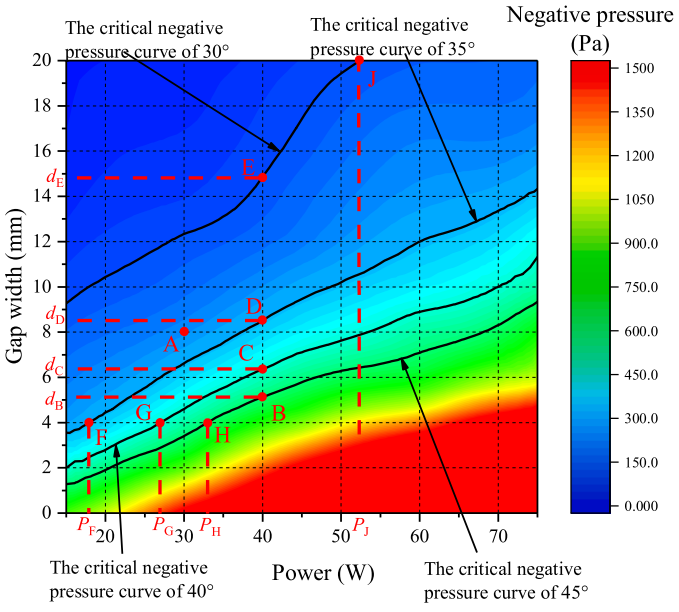


Fig. 10. Obstacle crossing performance of the water-free cleaning robot.

4.4. Analysis of safe operating conditions

The effect of the centrifugal fan power and gap width on the negative pressure is investigated. In PV power stations, lock blocks are used to attach the PV panels, resulting in a gap of 20 mm between the PV panels. The critical pressure Δp is obtained from Eq. (8). The working characteristics of the robot are shown in Fig. 10.

The horizontal axis represents the power P of the negative pressure adsorption system, and the vertical axis represents the gap width d during cleaning operations. The red dots (A, B..., J) represent the negative pressure inside the negative pressure cavity. The black lines represent the critical negative pressure curves at the same installation angle. At locations below the critical negative pressure curve, the cleaning robot can work safely on the PV panels with the given or smaller inclination angle.

The diagram can be used to determine the maximum gap between the PV panels that the cleaning robot can safely cross at different inclination angles and given power of the centrifugal fan. Point E corresponds to gap width d_E, which is the maximum gap width that the cleaning robot can cross at a 30° inclination angle. As the inclination angle decreases, the maximum gap width for safe crossing decreases. The maximum gap width is d_D at an inclination angle of 35°. When the gap width is known, the minimum power required for the cleaning robot to safely cross the gap can be obtained. Point F corresponds to power P_F of the centrifugal fan, which is the minimum power required when the cleaning robot safely crosses a gap with this width.

4.5. Performance evaluation

An experiment is conducted to evaluate the cleaning performance of the robot and the light transmittance and efficiency of the PV panels. The dust cleaning rate ξ_c, the increase rate τ_a of the transmittance, PV efficiency η_c, and the increase rate τ_c of the PV efficiency are used to evaluate the performance. These parameters are defined in Eqs. (10)–(13).

$$\xi_c = \frac{\beta_a}{\beta_c} \tag{10}$$

$$\tau_a = \frac{\beta_b - \beta_a}{\beta_b} \tag{11}$$

$$\eta_c = \frac{UI}{G_{sun} A_p} \tag{12}$$

$$\tau_c = \frac{\eta_{cb} - \eta_{ca}}{\eta_{cb}} \tag{13}$$

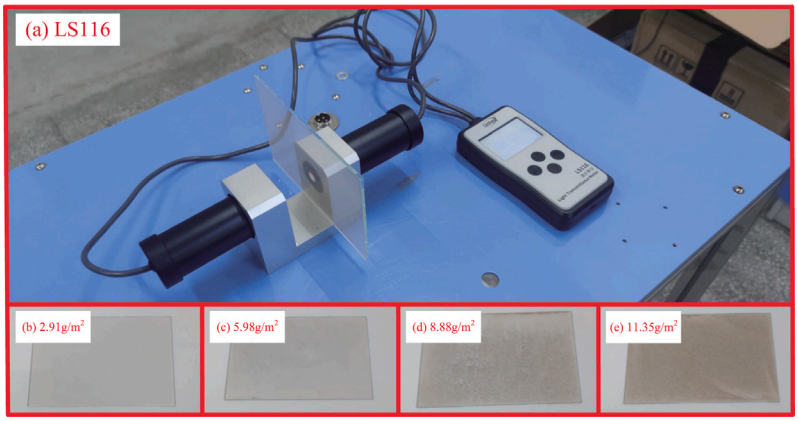


Fig. 11. Experimental equipment to measure the light transmittance.

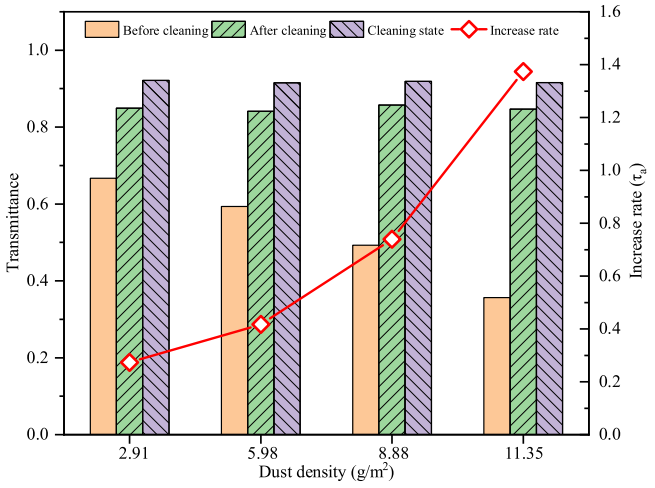


Fig. 12. The experimental results of the transmittance test.

where β_c, β_b, and β_a denote the transmittance of the clean panels and the panels before and after cleaning, respectively. U denotes the output voltage, I denotes the output current, G_{sun} denotes the irradiance, and A_p denotes the area of the PV panels. η_{cb} and η_{ca} are the PV efficiency before and after cleaning, respectively.

4.5.1. Transmittance

The test sample is ultra-white toughened glass with a size of 100 mm × 150 mm × 3.2 mm (transmittance test) and 30 mm × 100 mm × 3.2 mm (dust density calibration). The two samples are placed in an airtight box so that the dust can settle. The LS-116 transmittance tester is shown in Fig. 11(a), and the four glass samples with different dust levels are shown in Fig. 11(b)–(e).

The four glass samples with different dust densities are attached to a 30° PV panel, and the water-free cleaning robot is used for cleaning experiments. The LS116 transmittance tester is used to measure the transmittance before and after cleaning. The experimental results of the transmittance are shown in Fig. 12.

The transmittance has significantly improved after cleaning by the PV robot. The transmittance of the glass samples is 27.36%, 41.73%, 73.98%, and 137.35% higher after cleaning at dust densities of 2.91 g/m^2, 5.98 g/m^2, 8.88 g/m^2, and 11.35 g/m^2, respectively. After cleaning, the transmittance of the glass samples is lower than that of the clean panel. The reason is that there may be residual dust on the cleaned glass surface. It is difficult to remove small dust particles using the rolling brush and negative pressure dust removal system of the water-free cleaning robot. According to Eq. (10), the cleaning rate ξ_c of the four glass samples is 92.2%, 91.88%, 93.27%, and 92.48%, respectively. Thus, the average dust cleaning rate is 92.46%, meeting the design requirements (in Table 1).

4.5.2. PV efficiency

Three irradiation levels (high irradiance (> 600 W/m^2), medium irradiance (300 W/m^2 - 600 W/m^2), and low irradiance (<300 W/m^2)) and three angles (0°, 25°, 30°) are selected to evaluate the performance of the robot under actual operating conditions. A PROVA 1011 I-V tester is used to measure the voltage, current, and maximum power of the PV panels and the irradiance. The PV efficiency η_c and the increase rate τ_c of the PV efficiency before and after cleaning are obtained by Eqs. (12) and (13). The experiments are carried out on a 2-kW distributed PV system on the roof of a university in Northeast China. The results of the PV efficiency at three irradiance levels are shown in Fig. 13.

Fig. 13(a), (b), and (c) show that the PV efficiency after cleaning at high, medium, and low irradiance levels are 766 W/m^2, 414 W/m^2, and 235 W/m^2, respectively. The increase rate of the PV efficiency increases with the dust density, but is slightly affected by the irradiance and the angle of the PV panel. The average increase rates of the PV efficiency at high irradiance are 14.79%, 22.38%, and 27.53% at dust densities of 2.11 g/m^2, 5.61 g/m^2, and 10.44 g/m^2, respectively. At medium irradiance, the average increase rates of the PV efficiency are 16.16%, 35.22%, and 42.64% at dust densities of 2.45 g/m^2, 8.02 g/m^2, and 11.03 g/m^2, respectively. At low irradiance, the average increase rates of the PV efficiency are 17.66%, 26.78%, and 41.69% at dust densities of 33.17 g/m^2, 7.69 g/m^2, and 10.64 g/m^2, respectively. The PV efficiency is restored to the initial value after cleaning, but small fluctuations occur, which may be caused by a change in the instantaneous irradiance during the measurements. The experimental results show that the PV efficiency is improved after cleaning by the water-free robot for different dust densities, angles, and irradiance levels. The improvement in the PV efficiency ranges from 11.06% to 49.53%, verifying the effectiveness of the proposed robot.

4.5.3. Cleaning efficiency

Manual cleaning, EDS, superhydrophobic coating, super water jet, ultrasonic cleaning described in Kazem et al. (2020), and mechanical cleaning described in Hammoud et al. (2019) are conducted to compare the robot's cleaning performance. The experiment is carried out with two 30° PV panels with the same dust density on the surface. The experimental setup is shown in Fig. 14.

The photos show that the water-free cleaning robot can move autonomously on the surface of the PV panels owing to the negative pressure adsorption system and the wheeled travel system. It can cross

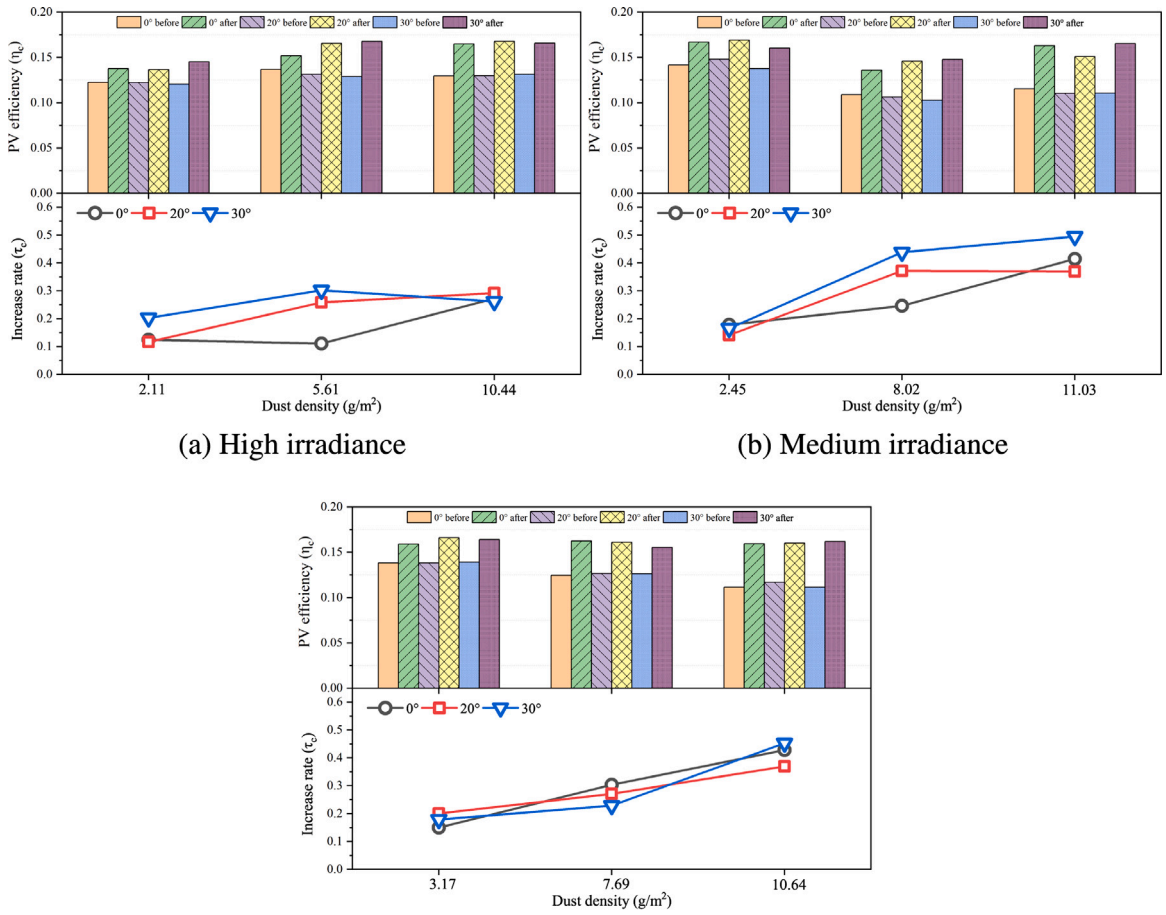


Fig. 13. The PV efficiency at three irradiance levels.

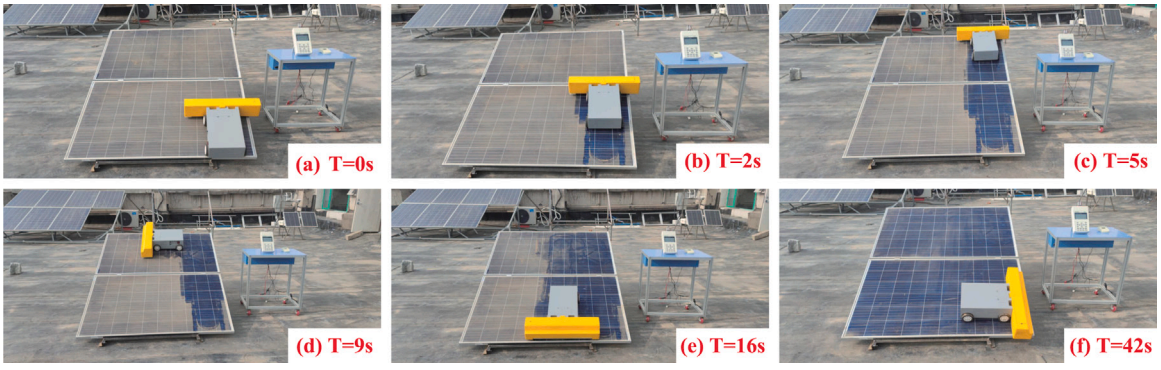


Fig. 14. The cleaning experiment of the water-free cleaning robot.

the 2 mm gap between the PV panels. The dust is collected by the rolling brush and negative pressure dust removal system, and the cleaning time is 42 s. The comparison of different cleaning methods are listed in Table 2.

Table 2 illustrates the comparison of different cleaning methods based on energy supply, operation medium, operation condition, operation cost, and field application. The comparison shows that EDS and ultrasonic cleaning are high energy consumption, high cost, and require special operation condition; thus, it is rarely used in the PV field. Superhydrophobic coating and super water jet have no moving parts and do not require human intervention. However, the efficiency of the coating will decrease over time. These methods are somewhat effective with rain, but in dry conditions regular washing should be used. Manual cleaning, mechanical cleaning and the proposed robot in this paper are further compared, and the results are listed in Table 3.

The manual cleaning method uses water. This method has the highest increase rate of the PV efficiency (36.33%) and the longest cleaning time (108 s). The mechanical cleaning method has the shortest cleaning time (7 s) and the lowest increase rate (18.77%). Because the cleaning equipment is heavy, the edge of the PV panels is used as a support. This system is often permanently installed. Distributed PV systems in water-scarce areas typically have a small installed capacity and scattered layout. The water-free cleaning robot can replace the manual cleaning method in high-altitude or dangerous areas. The proposed robot system has higher cleaning efficiency than manual cleaning, and the increase rate of the PV efficiency is 1.55 times higher than that of

Table 2
Comparison of different cleaning methods.

Methods	Energy supply	Operation			Field application
		Medium	Condition	Cost	
Manual cleaning (Kazem et al., 2020)	human intervention	water	–	"high"	✓
EDS (Kazem et al., 2020)	high voltage	–	dry	"high"	–
Superhydrophobic coating (Kazem et al., 2020)	–	–	humidity	"low"	✓
Super water jet (Kazem et al., 2020)	–	water	humidity	"low"	✓
Ultrasonic cleaning (Kazem et al., 2020)	specific power	water	specific	"high"	–
Mechanical cleaning (Hammoud et al., 2019)	DC power	–	–	"medium"	✓
Proposed robot	DC power	–	–	"low"	✓

Table 3
Comparison results of cleaning efficiency for three methods.

Methods	Installation	Increase rate τ_c	Cleaning time (s)	Weight (kg)
Manual cleaning (Kazem et al., 2020)	–	36.33%	108	–
Mechanical cleaning (Hammoud et al., 2019)	fixed	18.77%	7	⩽70
Proposed robot	portable	29.18%	42	⩽8

the mechanical cleaning method. The proposed robot also has a small volume and weight and is suitable for dust removal from panels in distributed PV systems in water-scarce areas.

5. Conclusions

A novel water-free cleaning robot was proposed for dust removal from PV panels in distributed PV power stations. A negative pressure adsorption and wheeled travel system and the rolling brush and negative pressure dust removal system were developed to ensure the stable operation of the robot. Three irradiance levels (high, medium, and low) and three installation angles (0°, 20°, and 30°) were considered to analyze the effectiveness of the negative pressure adsorption system and the obstacle crossing and cleaning abilities of the robot. The main conclusions are summarized as follows:

(1) The critical adsorption force, critical pressure, and critical power of the negative pressure adsorption system were determined using a mechanical model of the robot. The effectiveness of the negative pressure adsorption system was verified by experiments with load weights of 60, 70, and 80 N and inclination angles ranging from 0° to 30°.
(2) The gap width and height substantially affected the pressure of the negative pressure adsorption system. At a gap width of 2 mm and 30 W power of the centrifugal fan, the negative pressure decreased by 66.39% as the gap height increased from 0 mm to 10 mm.
(3) The effectiveness of the water-free cleaning robot was verified by evaluating the light transmittance of the panels and the PV efficiency. The results showed that the average dust cleaning rate was 92.46%.

Although the water-free cleaning robot can successfully remove dust, its cleaning efficiency requires improvement. In the future, we will optimize the number and configuration of the negative pressure adsorption system to improve the robot's obstacle and crawling ability. In addition, the dust removal system should have a modular design to be able to switch between brushes of different materials to adjust to different conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Key R&D Program of China Grant. 2018YFB1500800, Science and Technology Development Program of Jilin Province, China Grant. 20190302079GX, Science and Technology Project of State Grid Corporation of China Grant. SGTJDK00DYJS2000148.

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